

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                        | Marks Available |          |          |           |          |           |
|----------|-----|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                        | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
|          | (c) |  |  | Substitution: $0.35 = \frac{2.8}{R}$ (1)<br>Manipulation: $R = \frac{2.8}{0.35}$ (1)<br>Resistance = 8 000 [ $\Omega$ ] or 8 k[ $\Omega$ ] (1)<br>$R = 8 \times 10^n$ ( $n \neq 3$ ) $\rightarrow$ 2 marks<br>$2.8 \times 0.35 = 0.98 \rightarrow$ 0 marks [unless first mark awarded] | 1               | 1<br>1   |          | 3         | 3        |           |
|          |     |  |  | <b>Question 1 total</b>                                                                                                                                                                                                                                                                | <b>4</b>        | <b>6</b> | <b>5</b> | <b>15</b> | <b>9</b> | <b>12</b> |